EXP 3 ANALYSIS OF CELLULAR SPECTRUM AND PROPAGATION: SUB-1 GHZ, MID-BAND AND MMWAVE PATH LOSS AND INTERFERENCE

MATLAB CODE:

```
clc;
clear;
close all;
%% CELLULAR SPECTRUM AND PROPAGATION ANALYSIS
% Frequency bands
frequency = [700e6, 3.5e9, 28e9];
bandNames = {'Sub-1 GHz','Mid-Band','mmWave'};
% System parameters
distance = 100:100:2000;     % Distance in meters
Pt = 30;                     % Transmit power in dBm
Gt = 0;                      % Transmitter antenna gain in dBi
Gr = 0;                      % Receiver antenna gain in dBi
% Reference distance
d0 = 1;
%% 1. PATH LOSS CALCULATION
pathLoss = zeros(3,length(distance));
for i = 1:3
    lambda = 3e8/frequency(i);
    % Free Space Path Loss at reference distance
    PL0 = 20*log10(4*pi*d0/lambda);
    % Simplified propagation model
    pathLoss(i,:) = PL0 + 30*log10(distance/d0);
end
```

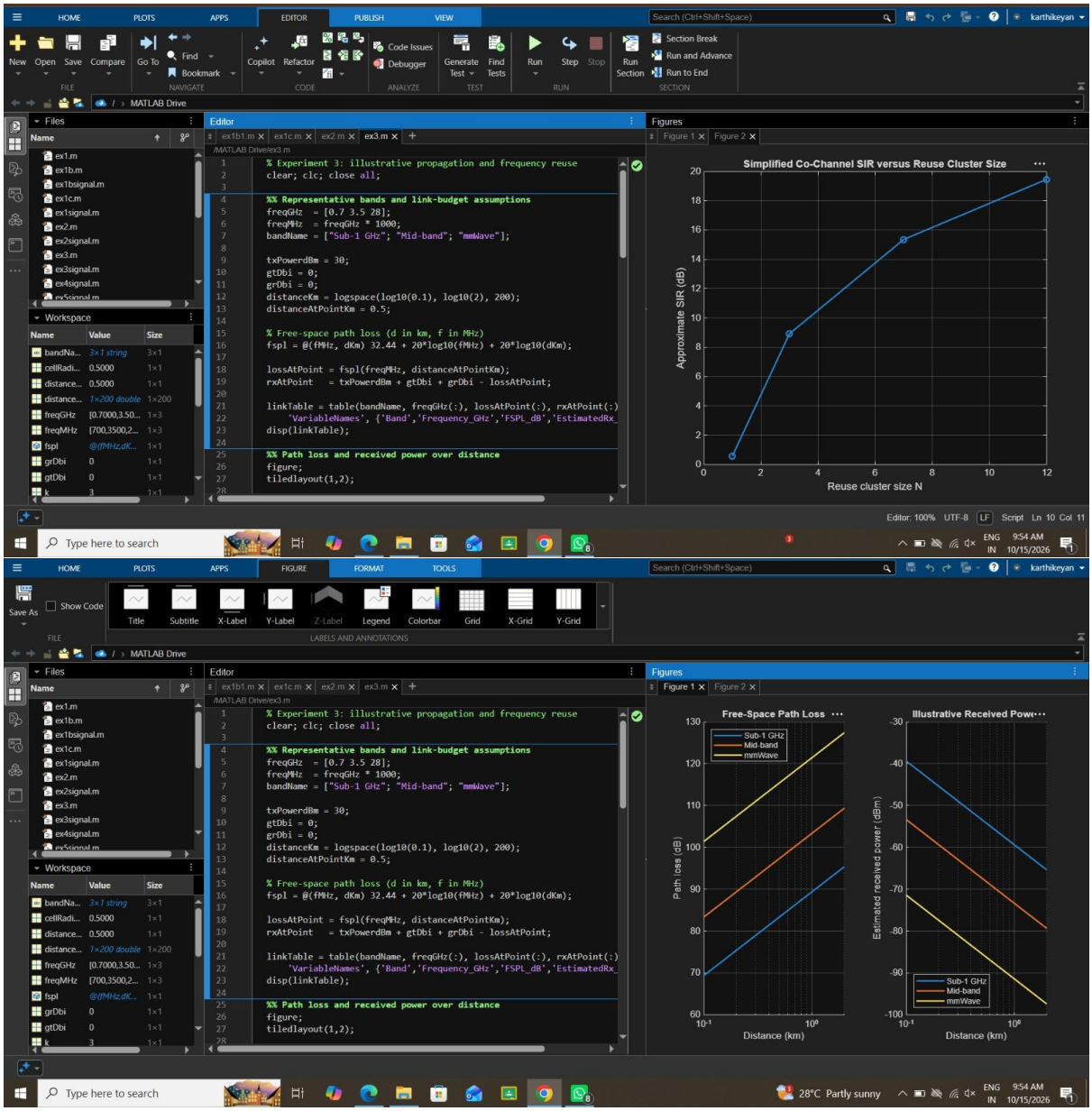
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%% 2. PLOT PATH LOSS COMPARISON
figure('Name','Path Loss Comparison');
plot(distance,pathLoss(1,:),'b','LineWidth',2);
hold on;
plot(distance,pathLoss(2,:),'r','LineWidth',2);
plot(distance,pathLoss(3,:),'g','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Path Loss (dB)');
title('Path Loss Comparison of Cellular Frequency Bands');
legend(bandNames,'Location','northwest');
%% 3. RECEIVED SIGNAL POWER
receivedPower = zeros(3,length(distance));
for i = 1:3
    receivedPower(i,:) = Pt + Gt + Gr - pathLoss(i,:);
end
figure('Name','Received Signal Power');
plot(distance,receivedPower(1,:),'b','LineWidth',2);
hold on;
plot(distance,receivedPower(2,:),'r','LineWidth',2);
plot(distance,receivedPower(3,:),'g','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Received Power (dBm)');
title('Received Signal Power vs Distance');
legend(bandNames,'Location','southwest');
%% 4. INTERFERENCE ANALYSIS
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% Three interfering base stations
interferenceDistance = 300;
interferencePower = zeros(3,length(distance));
for i = 1:3
    lambda = 3e8/frequency(i);
    PL_interference = 20*log10( ...
        4*pi*interferenceDistance/lambda) + ...
        30*log10(distance/d0);
    interferencePower(i,:) = Pt - PL_interference;
end
figure('Name','Interference Analysis');
plot(distance,interferencePower(1,:),'b','LineWidth',2);
hold on;
plot(distance,interferencePower(2,:),'r','LineWidth',2);
plot(distance,interferencePower(3,:),'g','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Interference Power (dBm)');
title('Interference Power Comparison');
legend(bandNames,'Location','southwest');
%% 5. SINR CALCULATION
noisePower = -100;       % Noise power in dBm
SINR = zeros(3,length(distance));
for i = 1:3
    signal_mW = 10.^(receivedPower(i,:)/10);
    interference_mW = 10.^(interferencePower(i,:)/10);
    noise_mW = 10^(noisePower/10);
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```
    SINR(i,:) = 10*log10( ...
        signal_mW ./ (interference_mW + noise_mW));
end
figure('Name','SINR Analysis');
plot(distance,SINR(1,:),'b','LineWidth',2);
hold on;
plot(distance,SINR(2,:),'r','LineWidth',2);
plot(distance,SINR(3,:),'g','LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('SINR (dB)');
title('SINR Comparison of Cellular Frequency Bands');
legend(bandNames,'Location','best');
%% 6. FREQUENCY COMPARISON
figure('Name','Frequency Band Comparison');
bar(frequency/1e9);
set(gca,'XTickLabel',bandNames);
ylabel('Frequency (GHz)');
title('Cellular Spectrum Frequency Bands');
grid on;
%% 7. PATH LOSS AT 1000 METERS
index = find(distance == 1000);
fprintf('\n--- CELLULAR PROPAGATION RESULTS ---\n');
for i = 1:3
    fprintf('\n%s\n',bandNames{i});
    fprintf('Frequency: %.2f GHz\n', ...
        frequency(i)/1e9);
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```
    fprintf('Path Loss: %.2f dB\n', ...
        pathLoss(i,index));
    fprintf('Received Power: %.2f dBm\n', ...
        receivedPower(i,index));
    fprintf('SINR: %.2f dB\n',SINR(i,index));
end
%% 8. FINAL COMPARISON TABLE
Results = table( ...
    bandNames', ...
    frequency'/1e9, ...
    pathLoss(:,index), ...
    receivedPower(:,index), ...
    SINR(:,index), ...
    'VariableNames', ...
    {'Frequency_Band','Frequency_GHz', ...
    'Path_Loss_dB','Received_Power_dBm','SINR_dB'});
disp(' ');
disp('--- FINAL COMPARISON TABLE ---');
disp(Results);
fprintf('\nSimulation Completed Successfully!\n');
```

OUTPUT:


Simplified Co-Channel SIR versus Reuse Cluster Size
Approximate SIR (dB)
Reuse cluster size N
Free-Space Path Loss
Illustrative Received Power
Path loss (dB)
Estimated received power (dBm)
Distance (km)
Sub-1 GHz
Mid-band
mmWave